



Kigali- Rwanda
Email : info@rtb.gov.rw
Web site : www.rtb.gov.rw
P.O Box 4940 Kigali

**Training Organization Guide (ToG)
For
TVET CERTIFICATE V in AUTOMOBILE TECHNOLOGY CURRICULUM**

Table of contents

Part 1: Pedagogical organisation	3
1. Presentation of the training programme	3
1.1. The qualification	3
1.2. Minimum Entry requirements	3
2. The training programmes	3
2.1 Information about competencies	3
3. Human resources	5
3.1. Trainers' qualifications and competences (Trainers' profile)	5
Part 2: Resources organisation	14
1. Material Resources	14
1.1. Category 1: Machines, equipment, and accessories	14
1.2. Category 2: Tools and instruments	26
1.3. Category 3: Security materials	37
1.4. Category 4: Consumables and raw materials	43
2. Training facilities	49
2.1. Furnitures and office materials	49
2.2. Audio-visual materials	54
2.3. Didactic materials	55
2.4. Infrastructures	56
Reference books:	59

Part 1: Pedagogical organisation

1. Presentation of the training programme

1.1. The qualification

This qualification provides the knowledge, skills and attitudes for a learner to be competent in a range of routine tasks and activities that require the application of a limited range of basic practical skills in a defined context. Work would be undertaken in various transport and logistics industries where Vehicle spare parts technician, Automotive electronic technician, Vehicle automatic gearbox technician, Diesel injection system technician, Hybrid vehicle technician are provided. Learners may work with minimum supervision.

1.2. Minimum Entry requirements

The minimum entry requirement to this qualification is to be physically and mentally capable with an interest in Automobile Technology and related areas. For candidates entering this qualification must completed TVET certificate level 4 in Automobile Technology and a range of other related TVET certificate level 4 in Automobile.

2. The training programs

2.1 Information about competencies

No	Co-Curricular activities	Credit
1	Sports/Clubs	1
2	Self-study/Library Reasearch	1
Total		2

	No	Code	Core competencies	Credit
GENERAL		CCMBO502	Organise a Business	3
		CCMCZ501	Develop Attitudes of Living Together In Harmony	3
		CCMEN502	Use Upper- Intermediate English at The Workplace	3
		CCMFT502	Echanger les Idées en Français Elementaire	3
		CCMIW502	Apply ICT at Workplace	3
		CCMKN502	Gukoresha Ikinyarwanda cy'intyoza	3
		CCMPE502	Apply Professional and Multi-Cultural Ethics at Workplace	3
		GENAC502	Apply Basics of Analytical Chemistry	4
		GENEP502	Apply Logarithms, Exponential Equations and Probability	6
		GENAP502	Apply Dynamics and Mechanical Waves	5
		AUTIA501	Integrate the Workplace	20
		GENHP501	Apply Basic Hydraulic and Pneumatic Systems	11
SPECIFIC		AUTDI501	Repair Diesel Injection System	12
		AUTAG501	Repair Vehicle Automatic Gearbox	12
		AUTES501	Repair Vehicle Electronic Systems	12
		AUTHV501	Repair Hybrid Vehicle	12
		AUTSM501	Manage Auto Spare Parts and Materials	5
	Total			120

3. Human resources

3.1. Trainers' qualifications and competences (Trainers' profile)

No.	Modules	Qualification	Field or Industry experience	Technical teaching experience	Languages skill	ICT skills.
1.	GENAM502/ Mathematics	-A0 in Mathematics with education A0 Mathematics -A1/Advanced Diploma in Mathematics with education - -A1/Advanced Diploma in Mathematics	No Field or Industry experience required	-No teaching experience required, -However, having any, experience in teaching will be an added advantage.	Fluent in English; French is an added advantage	Microsoft word Microsoft excel Microsoft power point Internet exploration Microsoft equations
2.	GENDM502/ Physics	-A0 in Physics with education -A0 in Physics -A1/Advanced Diploma in Physics with education	No Field or Industry experience required	-No teaching experience required, -However, having any, experience in	Fluent in English; French is an added advantage	Microsoft word Microsoft excel Microsoft power point Internet exploration

		-A1/Advanced Diploma in Physics		teaching will be an added advantage.		
3.	GENAC502/ Chemistry	-A0 in Chemistry with education -A0 in Chemistry -A1/Advanced Diploma in Chemistry with education -A1/Advanced Diploma in Chemistry	No Field or Industry experience required	-No teaching experience required, -However, having any, experience in teaching will be an added advantage.	Fluent in English; French is an added advantage	Microsoft word Microsoft excel Microsoft power point Internet exploration
4.	CCMEN502/ English	-A0 in English with Education - A0 in English -A1/Diploma in English with Education -A1/Advanced Diploma	No Field or Industry experience required	-No teaching experience required, -However, having any, experience in teaching will be	Fluent in English.	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration

		in English		an added advantage.		
5.	CCMFT502/ Français	-A0 in French with Education - A0 in French -A1/Diploma in French with Education -A1/Advanced Diploma in French	No Field or Industry experience required	-No teaching experience required, -However, having any, experience in teaching will be an added advantage.	Fluent French	Microsoft word Microsoft excel Microsoft power point Internet exploration
6.	CCMKK502/ Kiswahili	-A0 in Kiswahili with Education - A0 in Kiswahili -A1/Diploma in Kiswahili with Education -A1/Advanced Diploma in Kiswahili	No Field or Industry experience required	For any candidate with A0 or A1, no teaching experience required -However, having any experience in teaching -	Fluent in Kiswahili	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration

				Training on pedagogy (CBT&CBA) Will be an added advantage.		
7.	CCMKN502/ Ikinyarwanda	-A0 in Kinyarwanda with Education - A0 in Kinyarwanda -A1/Diploma in Kinyarwanda with Education -A1/Advanced Diploma in Kinyarwanada	No Field or Industry experience required	For any candidate with A0 or A1, no teaching experience required -However, having any experience in teaching - Training on pedagogy (CBT&CBA)	Fluent in Kinyarwanda	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration

				Will be an added advantage.		
8.	CCMPE502/ Professional Ethics		No Field or Industry experience required	For any candidate with A0 or A1, no teaching experience required -However, having any experience in teaching - Training on pedagogy (CBT&CBA) Will be an added advantage.	Fluent in English; French is an added advantage	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration

9.	CCMIW502/ ICT	A0 in ICT with Education - A0 in ICT -A1/Diploma in ICT with Education -A1/Advanced Diploma in ICT	No Field or Industry experience required	For any candidate with A0 or A1, no teaching experience required -However, having any experience in teaching - Training on pedagogy (CBT&CBA) Will be an added advantage	Fluent in English; French is an added advantage	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration - Software Developing - Computer Programming - Networking
----	---------------	---	--	---	---	---

10.	CCMCZ501 /Citizenship		No Field or Industry experience required	For any candidate with A0 or A1, no teaching experience required -However, having any experience in teaching - Training on pedagogy (CBT&CBA) Will be an added advantage	Fluent in English; French is an added advantage	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration
-----	--------------------------	--	---	---	---	--

11.	CCMB0502/ Entrepreneurship		No Field or Industry experience required	For any candidate with A0 or A1, no teaching experience required -However, having any experience in teaching - Training on pedagogy (CBT&CBA) Will be an added advantage	Fluent in English; French is an added advantage	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration
-----	-------------------------------	--	---	---	---	--

12-18	AUTIA501/ Industrial Attachment Program GENHP501/ Basic hydraulic and pneumatic systems AUTDI501/ Diesel injection system Repairing AUTAG501/ Vehicle Automatic Gearbox Repairing AUTES501/ Vehicle Electronic System Repairing AUTHV501/ Hybrid Vehicle Repairing AUTSM501/ Auto Spare Parts and Materials Management	- Bachelor of Technology in Mechanical Engineering/ Automobile Technology - Bachelor in Automotive and Power Engineering - Bachelor in Automobile Technology -A1/Advanced Diploma in Mechanical Engineering/Automobile Technology	No Field or Industry experience required	For any candidate with A0 or A1, no teaching experience required -However, having any experience in teaching - Training on pedagogy (CBT&CBA) Will be an added advantage.	Fluent in English; French is an added advantage	- Microsoft word - Microsoft excel - Microsoft power point - Internet exploration
-------	---	---	--	---	---	--

Part 2: Resources organisation

1. Material Resources

1.1. Category 1: Machines, equipment, and accessories Specification

SN	<i>Machines and Equipment</i>	<i>Specification</i>	Quantity	Unit price	Total	Type of room	Module code
1	Workshop crane	With 3wheels, Single acting cylinder Capacity =1ton extension range =925mm-1350mm hook's height to ground range =525mm-2295mm column height= 1600mm-1700mm overall base length =1500mm-1555mm chains and straps with accessories	2	2,500,000	5,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
2	Hand pallet truck	Capacity =1ton fork overall length range=1150mm-1200mm height Min/max:85/195(mm) wheel type:PU/RU/Nyron	2	650,000	1,300,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

3	Air Compressor	Tank capacity: 500Litres Power:5.5HP Voltage:380-400V Three phase Frequency:50HZ Designed pressure:10bars Volume of air flow:850L/min	1	4,160,000	4,160,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
4	Portable waste oil tank	Capacity: 20Gallons Air operated Three wheels Adjustable funnel height	2	400,000	800,000	Workshop	AUTDI501 AUTAG501 AUTHV501
5	Two stage hydraulic transmission jack	Load capacity: 1ton Lift range:870mm-1790mm	2	600,000	1,200,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
6	Jacks stands	Lock capacity: 2 tons Adjustable Dimensions: 13.4x8.7x7.4inchs	8	75,400	603,200	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

7	Motor vehicle with automatic gearbox	Engine power:169HP Fuel type: Petrol Torque:151lb-ft Torque rpm:4400rpm Engine displacement:2.0L Engine configuration:inline Number of cylinders:4 Transmission type: continuous variable transmission	1	32,000,000	32,000,000	Workshop	AUTAG501 AUTES501
8	Motor vehicle with automatic gearbox	Fuel type: Diesel Engine displacement:6.7L Transmission type: automatic gearbox with planetary gear set Engine power: 370 hp Torque: 850 lb-ft Payload Capacity: Up to 6,570 lbs Engine configuration: V type	1	80,000,000	80,000,000	Workshop	AUTAG501 AUTES501
9	Work bench	Material: steel Finish: steel Frame material:alloy steel Dimensions:25.5x78.75x34inches item weight: 85-90Kg	5	1,330,600	6,653,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

10	Vehicle lift	2-Post Hydraulic car lift Minimum lifting capacity 4000kg (4 tonnes) Electro-hydraulic controlled Power supply :380-400V Electric motor power:2.2KW Lifting height: 1900mm Total width:2820mm Working oil pressure: 18Mpa	2	4,500,000	9,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
11	Hydraulic press	Capacity: 30tons Item weight: 392Pound Product dimensions: 0.5x3.23x0.5inches Material: steel	1	4,500,000	4,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
12	Digital Multimeter	DC voltage: 200mV- 2/20/200/1000V, AC voltage: 2- 20/200/750V, AC current: 2- 20/200mA-20A, AC current: 2/200mA- 20A, Resistance: 200Ω- 2/20/200kΩ-2/20MΩ, Capacity: 2-20/200nF- 2/200μF, Temperature: - 20o +1000oC /-	5	85,000	425,000	Workshop	AUTDI501 AUTHV501 AUTES501

		4oF~+2000oF, Frequency: 20kHz, Power supply: 9V battery Accessories: multimeter battery charger, all probes, thermometer probes					
13	Scan tool	Free update Android 10, ROM 64GB, RAM 3GB Display resolution: 8" OBDI +OBDII Diagnosis Full set car connectors and cable kit. ECU Coding Injector coding All in one key programming Read and write most engines/gearboxes' ECUs without disassembling the shell. Multilanguage(English,Fr ench,Spanish,Germany,R ussian,Korea,Chinese,Ja panese,etc..) Open hidden features: make full use of hidden features or disable cars' annoying functions	1	3,500,000	3,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

		compatible for 12-24V Monitoring of STOP/START actuation. Reset: ABS Bleeding, Lights, injectors, dashboard, accelerator pedal					
14	Digital Clamp meter	Measuring Current Range: 40A/600A; Measuring Resistance Range: 400/4k/40k/400k/4M/4 0M; Measuring Inductance Range:10Hz~400kHz/400 kHz~4MHz; Dimensions:139.8*10.4*3 8mm; Max. Display:4000 counts	5	350,000	1,750,000	Workshop	AUTDI501 AUTHV501 AUTES501
	Diesel fuel injector nozzle ultrasonic cleaner	Mechanical power: 300W Ultrasonic cleaning power: 100W Simulation test speed range:10-990rpm Relative humidity: <85% Timing range:1-9999s Pulse width range: 0.51- 25ms Working volume: 3000ml Product size: 480x530x730mm	1	400,000	400,000	Workshop	AUTDI501

15	Common rail injector and pump test bench.	Supply Voltage: 380V(3P+N+GND) Motor power:50HP High pressure pump pressure:0-2200bar Low pressure:0.5-6Bar Speed:0-400rpm adjustable Dimension: (LXHXW:1350X1500X750 mm) Oil tank capacity: 20L Power output: 8KW	2	6,000,000	12,000,000	Workshop	AUTDI501
16	Injection pump tester	Speed adjustment range: ±4000rpm Number of cylinders that can be tested: 12 Spindle speed fluctuation: ±1rpm Motor power: 11KW Fuel tank capacity:60L Fuel pump supply: >10L/min Size:1700x960x1750mm	1	6,000,000	6,000,000	Workshop	AUTDI501
17	Engine hoist	Load Capacity : 20tons Lifting height: 1-6m Manually operated Four wheels Adjustable height	1	2,000,000	2,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

18	Hybrid/Electrical car Personal Protective Equipment	Insulated gloves: Must meet ASMT D120 standard, Rubber material, Class 0 Safety glasses: Must meet ANSI Z87.1 for high impact eye protection, polycarbonate material or similar impact resistant material. Arc flash clothing: Meet NFPA 70E, flame resistant fabric. Face shield: Must meet ANSI Z87 , a clear polycarbonate materials Safety shoes: Must meet ASMT F2413,Leather material	5	650,000	3,250,000	Workshop	AUTHV501
19	Battery Lifter	Weight: 82KG Capacity: 3000kg Fork pocket:200x85mm Maximum lift height:43 in Fork depth:43 in Height: 61 inch width: 31-1/2 in length of fork:42	2	15,000,000	30,000,000	Workshop	AUTHV501

20	Insulation resistance tester(Megohmmeter)	Selectable test voltage: 250V, 500V,1000V and 2500V Insulation resistance: up to 500GΩ Automatic PI and DAR calculations	2	2,150,000	4,300,000	Workshop	AUTHV501
21	Thermal imaging camera (IR camera)	Model: TG275, instructions, wrist strap, USB A to USB C cable Display: LCD Power source: Battery powered High resolution camera 19000Pixel IR Ultra Clear Images Wireless connectivity Laser pointer	2	1,200,000	2,400,000	Workshop	AUTHV501

22	Battery cell balancer	Power input: AC90-264/40-60HZ WFI and BT Power: 600W Max. Equalize number of channel:2x12Pin Charge and discharge voltage range: 1.8-4.2V Equalize interface: 16Pin Voltage detection accuracy: $\pm 0.1\%$ FS2 $\pm$ mv Battery interface: 24pin Display:7-inch TFT LCD 1024X600 screen resolution Current detection interface: $\pm 1\%$ FS+0.05A Communication interface: SMA, USB-Device Battery temperature detection accuracy: ± 2 (9-25-85)Celcius degree	1	8,000,000	8,000,000	Workshop	AUTHV501
23	Hybrid vehicle battery charger	Voltage Range: 200V to 800V Power Output: Up to 22 kW Weight: Approximately 20 kg (44 lbs) Charging Standard: CCS, CHAdeMO	1	10,000,000	10,000,000	Workshop	AUTHV501

		Cooling System: Air-cooled					
24	Workshop tool trolley	Dimension: 840x405x810mm Material: Steel frame with stainless steel top Wheels: 4 casters (2 locking) Load capacity: 50kg Overall height:1000mm Number of level:3	5	500,000	2,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
25	Battery storage rack	4 Shelves Made of durable tubular powder coated steel Minimum load capacity:300kg for every shelf	1	500,000	500,000	Workshop	AUTHV501
26	Electrical/Hybrid vehicle battery high voltage tester	Input Voltage Maximum voltage: 1000V Two poles Suitable for AC and DC Testing range: 24-1000V AC and 6-1000V	2	150,000	300,000	Workshop	AUTHV501
27	Bench vice	Jaw opening: 200mm Jaw width : 240mm Mounting: screw Jaw depth:98mm	5	800,000	4,000,000		AUTDI501 AUTAG501 AUTHV501 AUTES501

28	Drilling machine	Power input: 600W No load speed: 2600rpm Power output: 360 Rated speed: 0-1676rpm Rated torque: 20Nm Chuck capacity: 1.5-13mm Drilling capacity: Alumium 13mm, wood 30mm,steek 30mm. Reverse direction.	1	400,000	400,000		AUTDI501 AUTAG501 AUTHV501 AUTES501
29	Cleaning tank	Internal tank dimension: 710x480x200mm Tank capacity: 300L Pump: 230V 50Hz, 26W Pump capacity: 10-12L/min	3	400,000	1,200,000		AUTDI501 AUTAG501 AUTHV501 AUTES501
30	Magnetic portable LED Work Light	Light Source Type LED Power Source: rechargeable (220V)	5	50,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
	SUB TOTAL 1				238,391,200		

1.2. Category 2: Tools and instruments

SN	Tools	Specification	Quantity	Unit price	Total	Type of room	Module code
1	Grease gun	Die-cast aluminium head, Ergonomic grip for ease of access to tough greasing points, Corrosion-resistant brass valve, Takes 400 g cartridges or bulk grease up to 500 cc, Tube size: 54 (interior) x 260 mm Air inlet: 1/4" GAS, Operating pressure: 4÷10 bar	2	150,000	300,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

2	Heavy Duty Metal Box(Tool Set with 3-Drawer)	Containing item numbers: Ring spanners: 6-32mm Open spanners:6-32mm Combination spanners:6-32mm Socket spanners:6-32mm T-Bar Extension bar Universal joint Ratchet Set of flat screw drivers Set of Philips screw driver (star) Set of pliers(universal, long nozzle, cutting, combined, multi-grip, locking plier and Wire stripping pliers) Torx key set Allen key set Filler gauge Wire cutting knife Hammer Soft hammer Spark plug spanner :16mm Hand riveter Hacksaw with brades circlip pliers (external & internal) Pick and hook sets Pick up tools	1	1,500,000	1,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
---	--	--	---	-----------	-----------	-----------------	--

3	Measuring tape	Blade length:5 m, Graduation: metric/inches, Blade material: stainless steel, Cover material : Plastic /Stainless steel	25	6,000	150,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
4	Slage hammers	Weight: 5kg Length :900mm	2	35,000	70,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
5	Hack saw	Blade Material : Metal Handle: Soft grip handle, Plastic coated, High blade tensioning mechanism Cutting angle : up to 90 Degrees, Blade Length : minimum 30cm	3	10,000	30,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
6	Torque wrench (Digital)	Item Dimension(LxWxH): 1.5 x 1.5 x 17 inches Material Metal, Item Torque 200 Nm	2	50,000	100,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
7	Swivel-end socket	The slidable T-handle is fitted with an end-stop to ensure it does not slide out.With swivel jointed head (torsion drive 50°, swing drive up to 90°). The swivel angle is secured by an integral spring. Material: Chrome vanadium steel, chrome plated. Content table use: 1 each socket	2	40,000	80,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

		spanner: Size 8; 10; 11; 13; 14; 17; 19					
8	Vernier caliper	ANALOGUE: Measuring range : 0-300mm/0-12"; Graduation : 0.05mm/ 1/128", Jaw length : 64mm Other factures Four usages for measuring outside diameter, inside diameter step and depth, Made of solid stainless steel, Distinct lines and figures etched against satin chrome finish, With locking screw for stable reading, DIGITAL: Material: Stainless Steel, Measurement Range: 0-150mm; Accuracy 0.01	25	200,000	5,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
9	Automatic center punch	Heavy Duty Center Punch Dimensions : 4-7/8" Long x 1/2" Diameter Point Diameter : 0.055 Inches (1.4 mm) Material Tool Steel Pattern Knurled Body	5	50,000	250,000		AUTDI501 AUTAG501 AUTHV501 AUTES501
10	Pen scribe	Material: carbon steel Length: minimum 250mm Weight: minimum 76g Length: minimum 127mm	15	35,000	525,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

11	Micrometer screw gauge (Internal, External and depth)	Outside digital /analog micrometer: Inch/metric, ratchet Stop, 0-1" range Inch/metric, ratchet Stop, 1"-2" range Inch/metric, ratchet Stop, 2"-3" range Inside digital/analog micrometer: Inch/metric, ratchet stop, 5-30mm, accuracy 0.01 Inch/metric, ratchet stop, 25-50mm, accuracy 0.01 Inch/metric, ratchet stop, 50-75mm, accuracy 0.01 Inch/metric, ratchet stop, 75-100mm, accuracy 0.01 Depth micrometer set: gauge range 0-100mm Ratchet screw gauge	5	200,000	1,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
12	Steel ruler	Material: Stainless steel Length: 500mm	5	50,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

13	Taps & die set (metric)	Taps: Cutting Diameter: minimum 3/8" (10mm); Shank Diameter.: minimum 1/4" (6mm); Dimensions : 5.51 x 3.66 x 1.06 inches Material: High-quality heat-treated tungsten carbide Shape: Round Material: High-Speed Steel Die: Item Weight: minimum 20 g; Wire pitch (mm): minimum 0.5 Drill length (mm): minimum 5; Total length (mm): minimum 54; Shank Diameter: minimum 1/4"; Dimensions: minimum 2.3 x 0.39 x 2.3 inches Cutting Diameter: minimum 10 Millimeters; Assembled Diameter: minimum 10 Millimeters Die and tape holders	2	150,000	300,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
14	Taps & die set (withworth)	Standard: 4", 6", 8", 10", 12", 1/4", 5/16", 3/8", 7/16", 1/2" Die and tape holders	2	150,000	300,000		AUTDI501 AUTAG501 AUTHV501 AUTES501

15	Set of punches	Materials : steel 24200 1/4-Inch Reg Center Punch, 24300 5/16-Inch Center Punch, 24301 3/8-Inch Center Punch, 24301 1/2-Inch Center Punch Chisels Punches	5	60,000	300,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
16	Filter wrenches	Wrenches are designed for high torque requirements of truck filters Designed for use with a 1/2" square drive tool 1-1/2" wide bands for added strength and increased grip area Belt type	5	60,000	300,000	Workshop	AUTDI501 AUTHV501
17	Drill bits set	Drill Bit Set (2-13mm), M35 HSS Metal Drill Bits for Steel, Stainless Steel, Hard Metal, Cast Iron, Plastic and Wood	5	45,000	225,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
18	Wheel spanner	Cross Wheel Spanner, Size : Material: Carbon steel	3	20,000	60,000	Workshop	AUTAG501 AUTHV501 AUTES501
19	Adjustable wrenches	Chrome Vanadium Steel, Item Length :24 Inches	5	50,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

20	Torque multiplier	Torque multiplier for right-hand and left-hand tightening, ratio 64: 1 Torque: 4800Nm Head style: Fixed square, hex	2	350,000	700,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
21	T-handle wrenches with swivelling hexagon sockets	Reliable, precise pre-positioning is the secret for the success of this wrench. The articulated joint maintains its function over the years, no adjustment being required. Socket rotation angle: 60°. Can be used at 90° in fixed position. Sliding T-handle, can be stopped in the required position throughout its length. Triple size marking on head.	5	30,000	150,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
22	Magnetic pick up tool	Flexible magnetic pick up tool bendable: Dimension:22.8" Pull force: 20Pound	5	15,000	75,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
23	Oil gun	Specifications: Material: Carbon steel & Aluminum alloy; Fluid Capacity: 250ml/250cc	5	15,000	75,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
24	Hydrometer	Product name: Battery Tester Hydrometer; Packaging material:single blister + color	5	7,000	35,000	Workshop	AUTHV501 AUTES501

		card; measurement accuracy:0.005					
25	Lamp tester	Type: Voltage Circuit Tester; Voltage: Standard; Power: 3W; Applicable Models: Universal; Name: automotive circuit voltage tester pen	10	18,000	180,000	Workshop	AUTHV501 AUTES501
26	Soldering station	Power Source: 230V Frequency: 50HZ Wattage: 600W Continuously adjustable temperature of hot air in the range of 100-500celsius degree	5	150,000	750,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
27	Dial gauge	Dial Gauge with Magnetic Stand ,Model: 94838, Material: Steel, Specifications: 0-10 mm 0.01 LC	5	50,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
28	Injection nozzle test rig	Pressure gauge range: 0-16Mpa Overall dimensions:150x120x350mm Capacity of fuel tannk: 3.5L Hand operated	1	300,000	300,000	Workshop	AUTDI501
29	C-clamps	Material: Malleable Steel, Size: 1 2 3 4 5 6 8 10 12 inches	5	20,000	100,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

30	Spark plugs cleaner and tester	Operating Voltage: 220V AC Recommended air pressure: 100 to 140 psi (7 to 10kg/cm2) Cleaner section abrasive capacity: 1 lb (453g) High-voltage output when spark testing : 30kV (30,000 V) Dimensions: 19.3" (49cm) x 11" (28cm) x 10.4" (26cm) Accessories included: 14mm-18mm spark plug hole adapters (2pcs each); 12mm, 14mm and 18mm cleaner gaskets (1 each); Adapter case (1pcs); Aviation Massive Electrode adapter plug (1pcs), Abrasive 1 lb (453g)	2	900,000	1,800,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
31	Files set	Recommended air pressure: 100 to 140 psi (7 to 10kg/cm2)	10	10,000	100,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
32	Puller set	Dimensions: 19.3" (49cm) x 11" (28cm) x 10.4" (26cm)	2	40,000	80,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

33	Bearing remover set	Weight: 22.5 lbs (10.2kg)	1	250,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
34	Hydraulic Floor jack	Accessories included: 12mm, 14mm and 18mm spark plug adapters (2pcs each); 12mm, 14mm and 18mm cleaner gaskets (1 each); Adapter case (1pcs); Aviation Massive Electrode adapter plug (1pcs), Abrasive 1 lb (453g)	1	1,500,000	1,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
35	Pry bar	Material:steel Length: 18"	10	20,000	200,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
36	Gasket scraper	Materai:steel Brade material: alloy steel Dimensions:8x1x0.25" Size: 8"	10	18,000	180,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
37	Air impact spanner	1 Hand operation forward and reverse Capacity bolt size: 16mm,square drive head 13mm Air inlet: 1/4" Air pressure: 90PSI Torque:678Nm	5	250,000	1,250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

38	Blow gun	Material: polyamide composite Air consumption: 14Nm ³ /h Noise level: 80dB Dynamic pressure:2.5bar	5	45,000	225,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
39	Thread pitch gauge	Type: metric and unified screw pitch gauge	5	30,000	150,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
	SUBTOTAL 2				19,340,000		

1.3. Category 3: Security materials

SN	Security materials	Specification	Quantity	Unit price	Total	Type of room	Module code
1	Goggles	Eyewear type: The goggles should primarily provide protection against impact & optical radiation and splashes. Weight: Should weigh approx 51g Widest lens: Should have ellipsoidal lens that extends the viewing range further, even peripherally. Comfort: Should have a soft PVC shroud that gently	25	10,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

		hugs the face and is indirect vented to allow cooling airflow and help prevent fogging.					
2	Mask	Respirator Rating Number: Class 95 Capacity: Removes 95% of all particles that are at least 0.3 microns in diameter.	200	2,000	400,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
3	Ear muffler (Ear plug)	Reduce noise from 35dB Adjustable and customised fit	100	20,000	2,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
4	Hand Gloves	Material: Latex Size: Small, Medium, Large	100	15,000	1,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
5	Safety helmet	Material: High Density Polyethylene Size: Fully adjustable internal harness and nape strap for extra comfort Shock absorption:	25	10,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

		vertically penetration resistant against sharp and pointed objects; Insulation level: flame resistance with an electrical insulation of 440 V					
6	Overcoats for trainers	Size: Medium, Large and Extra Material: 100% cotton, with long sleeve Size: Medium, Large and Extra	25	15,000	375,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
7	Overall for trainees	Type: Heavy Duty Material: 100% cotton Pockets: Poly Zip and Velcro Fasten two side Pocket, One back Pocket two breast pocket Size: Small, Medium, Large	25	20,000	500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

8	Safety shoes	1 4-Liter dry bag 1 Sewn bag Adhesive Plastic Bandages 1" x 3" - 20 pcs Assorted plasters (band-aid) pack of 20 - 1 pack Burn Gel Sachet 0.9g - 5 pcs Alcohol wipes - 10 pcs Conforming gauze roll 4" x 5yds - 2 pcs Foil blanket 160cm x 210cm - 1 pc Sterile gauze pads 4" x 4" 2pcs/pouch - 3 pouches Non-adherent pad 2" x 3" 1pc/pouch sterile - 1 pc Stainless steel tweezer - 1 pc Safety pins 12pcs/pack - 1 pack Skin closure strips H form 3.8cm x 7.6cm - 10 pcs Splint 4.25" x 24" - 1 pc Examination vinyl gloves non-sterile - 1 pair Tough-cut scissors 12cm -	25	120,000	3,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
---	--------------	---	----	---------	-----------	-----------------	--

		1 pc Trauma pad 5"X9" - 1 pc Triangular bandage 40" x 40" x 56" - 1 pc Wooden tongue depressor - 2 pcs Zinc oxide tape 2.5cm x 4.5cm - 2 pcs Emergency trauma bandage 6" - 1 pc Cohesive bandage 10cm x 4cm - 1 pc					
9	First Aid kit	With assorted medication for burns and cuts: e.g.: scissors, antiseptic cream, iodine, eyewash, bandages of 2 sizes, plaster, cotton wool; 1x Leaflet; 60 x Assorted plasters; 8 x Non-woven bandages; 16 x Dressing pads; 12 x Safety pins; 20 x Wipes; 6 x Eye pads; 3 x Pairs of gloves	5	120,000	600,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

10	Dry chemical extinguishers for class ABC	Test pressure: 27 bar Safety pin: steel Weight: 5kg	10	35,000	350,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
11	CO2 Extinguishers	Weight: 5kg Working Pressure: 150bar Hose: PVC hose with aluminum nozzle	10	35,000	350,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
12	Workshop directional signs kit	Prohibition signs Warning signs Mandatory signs Emergency signs	5	50,000	250,000		AUTDI501 AUTAG501 AUTHV501 AUTES501
13	Fire blanket	Material: Fiberglass, Working Temperature: Up to 1022°F (550°C) Product Size: 62"-84"	10	120,000	1,200,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
	SUB-TOTAL 3				11,025,000		

1.4. Category 4: Consumables and raw materials

SN	Consumables / materials	Specification	Quantity	Unit price	Total	Type of room	Module code
1	Fuel	Petrol and diesel	400	2,000	800,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
2	Cutting disc	Specification A 30 S BF, Diameter mm 115, Bore diameter mm 22.23, Thickness mm 2.5	100	3,500	350,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
3	Grinding disc	Disc øX Disc Width (mm): * 125 X 6	100	3,500	350,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
4	Alcohol	Isopropyl	50	10,000	500,000		AUTDI501 AUTAG501 AUTHV501 AUTES501
5	Soap	Powder soap Liquid soap	200	4,500	900,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
6	Engine degreaser	Sprayable Rinse off	50	10,000	500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

7	Silicone	Waterproof Sealant Single Tube of 85g	50	3,500	175,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
8	Insulating tapes	Insulating tape 50 mm wide, Soft and flexible PVC tape, Slightly embossed structure, Strong hold For all purposes 25 m roll	100	1,200	120,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
9	Automotive electrical harness	Full hybrid vehicle harness	5	500,000	2,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
10	Soldering wire	Diameter: 0.5-1mm Roll size: 55x50mm Unit: Roll weight: 250-300g	25	10,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
11	Starter battery	12V/50Ah Free maintenance Lead acid battery	5	90,000	450,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
12	Relays	Type: Make relay, changeover relay and Break relay Rated Current: 30A/40A; Terminal Type: 4Pin/5Pin; Coil Pick-up Volt.≤8V/16V;	50	5,000	250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

		Coil Res: *80Ω/320Ω; Coil Max. Voltage: 15.7VDC/31.4VDC (85°C); Voltage:12V/24V; Contact Capacity: 40A/14VDC 30A/14VDC; Coil Release Volt. ≥1.2V/2.4V (VDC); Coil Max. Voltage: 20.2VDC/40.5VDC (23°C)					
13	Vehicle fuse set	Breaking Capacity rating current:5A-40A Product Type:Blade Fuse Rating Voltage:12V Number of items per set: 140pcs	10	15,000	150,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
14	Automotive pin connector set	Connector type: Butt splice,crimp,spade,ring terminal,quick disconnect. Wire size range: 1.5-2.5mm ² Insulation material: Polyvinyl chrolide Material: copper,tin plate Number of item per set: 450Pcs	5	20,000	100,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
15	Battery jumper cable	Size: 3.5m Pair	5	20,000	100,000		AUTDI501 AUTAG501 AUTHV501 AUTES501

16	Ignition switch	Item dimension: 3.63X4.88X8" Voltage: 12V	5	250,000	1,250,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
17	Spark plugs set	Core material: Iridium Top material: Nickel Spanner size: 16mm Number of items per set: 4	5	60,000	300,000	Workshop	AUTHV501 AUTES501 AUTAG501
18	ECU	Engine and Transmission control unit	5	1,000,000	5,000,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
19	Sand paper	Grade range: 60-1500 Dimensions: 230x280mm	100	1,500	150,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
20	Shop rags	Industrial cleaning cloth Matherials: 100%cotton	100	15,000	1,500,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
21	Multimeter battery	Voltage: 9V Amperage: 500-650mAh Type: Rechargeable Lithium battery	50	12,000	600,000	Workshop	AUTHV501 AUTES501
22	Automatic Transmission Fluid	ATFDX III	50	8,500	425,000	Workshop	AUTAG501

23	Automatic gearbox gasket kit	Must OEM Standards Full kit	5	100,000	500,000	Workshop	AUTAG501
24	Anti-rust	WD-40	50	12,000	600,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
25	Reflective high voltage signal sign	Size: 14"x10" Material: reflective	10	25,000	250,000	Workshop	AUTHV501
26	Cable ties	Size: 15" Number of items per bag: 100	5	10,000	50,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
27	High-voltage insulation tape	Material PVC Application High Voltage Rated Voltage 600V Tensile Strength 16N/cm Min Thickness 7mil Temperature resistance 80°C	10	10,000	100,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
28	Electrical Conductive grease	Conductivity: <20Ohm.cm Applications: Maintain electrical continuity between moving, irregular, corroded or pitted surfaces and dissipates static electricity. Volume: 500ml	5	75,000	375,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501

29	Injector leak off washers set	Fitting position: On injector below leak off pipe Unit of scale: set of 4	5	20,000	100,000	Workshop	AUTDI501
30	Diesel Injector washers set	Material: copper Injector rig thickness: 3mm Number of pieces per set: 500	5	120,000	600,000	Workshop	AUTDI501
31	Injector nozzles	Material: High speed steel Package: 6 pcs Size: standard size	5	270,000	1,350,000	Workshop	AUTDI501
32	Diesel fuel filters	Depends on the type of vehicle	5	35,000	175,000	Workshop	AUTDI501
33	Diesel Injection pump gasket kit	Depends on the type of vehicle	5	25,000	125,000	Workshop	AUTDI501
34	Repair manuals	Depends on the type of vehicle	5	15,000	75,000	Workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
35	Funnel	Material Polypropylene Capacity 4.5 litres	5	15,000	75,000	Workshop	AUTDI501 AUTAG501 AUTHV501
	SUB -TOTAL 4				21,095,000		

2. Training facilities

2.1. Furnitures and office materials

SN	<i>Furniture and office materials</i>	<i>Specification</i>	Quantity	Unit price	Total	Type of room	Module code
1	Office Cupboard	Two-door steel locker, Size:H1850 x W900 x D450mm, Thickness:1.0mm, Volume:0.178cbm, Fitting:4shelves	5	1,200,000	6,000,000	Store/ Workshop	
2	Glass Cupboard/Bookcase Cabinet Unit	The Top Shelf to be fitted with double lockable glass door. The unit to be made of MDF Board 18mm thickness for the frame; Three Shelves. The back part on the unit should be reinforced with 18mm MDF board for rigidity and strength. Approx. dimensions of approx. 900(l) x 440(w) x 1500(h) mm high. The Lower section (cupboard) double lockable MDF board door. The unit to be made of MDF Board 18mm thickness for the frame;	1	1,200,000	1,200,000	Trainer's Office	

		Two Shelves. The back part on the unit should be reinforced with 18mm MDF board for rigidity and strength. Approx. Dimensions of approx. 900(l) x 440(w) x 900(h)mm high					
3	Heavy steel shelves for equipment storage	Size:47" x16" x 59" (L x W x H), 5 Layer full steel, Cold – Rolled steel plate	5	650,000	3,250,000	Trainer's Office	
4	Medium back officer Chairs	Passive ergonomic seating with built in lumbar support, Upright-locking position, Pneumatic seat, Height adjustment, Adjustable tilt tension control, colour: Black, Dimensions: .Back: 24"(H) x 20"(W) .Seat: 20"(D) x 20.5"(W) .Seat Height: 18"-21"	5	350,000	1,750,000	Trainer's Office	
5	Office Tables with drawers	Melamine faced chipboard, Desktop and side panel: 25mm, Modesty panel: 18mm, Color: beech, mahogany, Size: 1,200 x 600 x 750mm, Three fully	5	250,000	1,250,000	Trainer's Office	

		locking drawers on the right-hand side					
6	Waiting chair	Material:stainless steel Number of seats: 5 Beam and bracket material: Aluminum casting bracket Size: 3000x690x790mm	3	1,000,000	3,000,000	Trainer's Office	
7	Trainees Table	Size:Table: 1500x600x750mm, Chair: 360x360x440mm, Material: Plywood Fireproof board& 1.2mm Thickness Tube, 22mm Tube Diameter; Board: Fireproofing, scratch-resistant, without contaminated; Metal: Epoxy Powder coated, antirust, damp proof, strong	13	80,000	1,040,000	Classroom	
8	Table for Computer Room	Size:Table: 1500x600x750mm; Chair: 360x360x440mm; Material: Plywood Fireproof board& 1.2mm Thickness Tube, 22mm Tube Diameter Board: Fireproofing, scratch-resistant, without	13	80,000	1,040,000	ICT LAB	

		contaminated; Metal: Epoxy Powder coated, antirust, damp proof, strong					
9	Trainees Chairs	Size: L46"xW44"xH3" Armrest: N/A Material: 1.2mm durable steel pipe. Loading capacity: 200kg Back and seat: Solid wood Seating design: Bending smooth arc	25	75,000	1,875,000	Workshop	
10	Portable, Reversible Dry Erase Boards & Chalkboards	Size: 1800x1200mm Economy non-magnetic chalkboard on 2 sides Aluminium framed Free-standing Whiteboard on one side and black/blue chalkboard on the other side. Moves on four lockable wheels	5	350,000	1,750,000	Classroom, LAB, and Workshop	
11	Papers	A4 Cartons	10	8,000	80,000	Trainer's Office	
12	Pens	Blue Bic; Red Bic	100	300	30,000	Trainer's Office	
13	Stapling Machine	Rapid K2, 105 mm Length	5	10,000	50,000	Trainer's Office	

14	Staplers	24/6	10	2,000	20,000	Trainer's Office	
15	Paper clips	25mm, small Size; 33mm, medium Size; 50mm, Big sizes	100	1,000	100,000	Workshop	
16	Punching machines	8Cm, Punching Capacity: 36mm with calibrated Guide	5	10,000	50,000	Trainer's Office	
17	Stapler removers	Colour: black; Dimensions: 4.8X4X1.4 inches	5	2,000	10,000	Trainer's Office	
18	Pencils	Dolphin Super B Writer	20	500	10,000	Trainer's Office	
19	Notebooks	Note book	30	1,000	30,000	Trainer's Office	
20	Plastic files	A4	50	700	35,000	Trainer's Office	
21	Register books	A4	20	2,500	50,000	Trainer's Office	
22	Office trays	Tray size of 3 or 4, Size: 385X285X65mm	10	10,000	100,000	Trainer's Office	
23	Post it	3" x 5"/7.6 x 12.7 Cm, 12 Pads	20	1,000	20,000	Trainer's Office	
24	Non-permanent Markers	Dimension: D1.6 x L13.5 Cm; Fiber tips and writing smoothly; Quick drying; Good performance; non-Toxic and environmentally safe; Smell favourably	200	1,000	200,000	Classroom, LAB, and Workshop	

25	Chalks (Cartons of 36 Packets for each carton)	Cartons of white colours; Cartons of Colours	250	4,000	1,000,000	Classroom, LAB, and Workshop	
	SUB-TOTAL 5				23,940,000		

2.2. Audio-visual materials

SN	Audio-visual materials	Specification	Quantity	Unit price	Total	Type of room	Module code
1	Projectors and Accessories (Remote, White board,)	Clear Visuals: SVGA native resolution (800x600) for sharp views; Flexible Connectivity: Dual VGA-In connectors for easy switching between multiple presenters; DLP® Projector: Filter-free to reduce maintenance cost and utilize Brilliant Color TM for accurate, true to life images, DLP (Digital Light Processing); Native Aspect ratio: 1.25:1; Video Compatibility: Component, S-video, composite; HDTV: 720p, 1080i, 1080p/60, 1080p/24, 576i, 576p; Lamp Life: 5000 hours, 6000 eco-modes. Screen	4	800,000	3,200,000	ICT lab and Classroom	AUTDI501 AUTAG501 AUTHV501 AUTES501

		size: 60-100 inches; Portable Indoor and Outdoor projector.					
2	Computer	Laptop i7 RAM:8GB Processor:1.70 Hard disk: SSD 1TB	10	1,500,000	15,000,000	ICT LAB and Trainer's office	AUTDI501 AUTAG501 AUTHV501 AUTES501
3	USB discs	8GB	15	10,000	150,000	ICT LAB	AUTDI501 AUTAG501 AUTHV501 AUTES501
4	Printer	Laserjet pro MFP4103 Functions: Print, copy, scan, fax Print speed black (ISO, letter) Up to 42 ppm	2	600,000	1,200,000	Trainer's office	AUTDI501 AUTAG501 AUTHV501 AUTES501
	<i>SUB-TOTAL 6</i>				19,550,000		

2.3. Didactic materials

SN	<i>Didactic materials</i>	<i>Specification</i>	Quantity	Unit price	Total price	Type of room	Module code
1	Textbook lessons	Per Module	100	100,000	10,000,000	Library	AUTDI501 AUTAG501 AUTHV501 AUTES501

2	Posters	Safety instructions, Technical information, Regulatory compliance, Education content, Customer information. Dimension: A2 (594mm x 420mm), A3 (420mm x 297mm) and A4 (297mm x 210mm) Material: Fabric	20	20,000	400,000	Classroom, laboratory, or workshop	AUTDI501 AUTAG501 AUTHV501 AUTES501
	SUB-TOTAL 7				10,400,000		
	General Total				343,741,200		

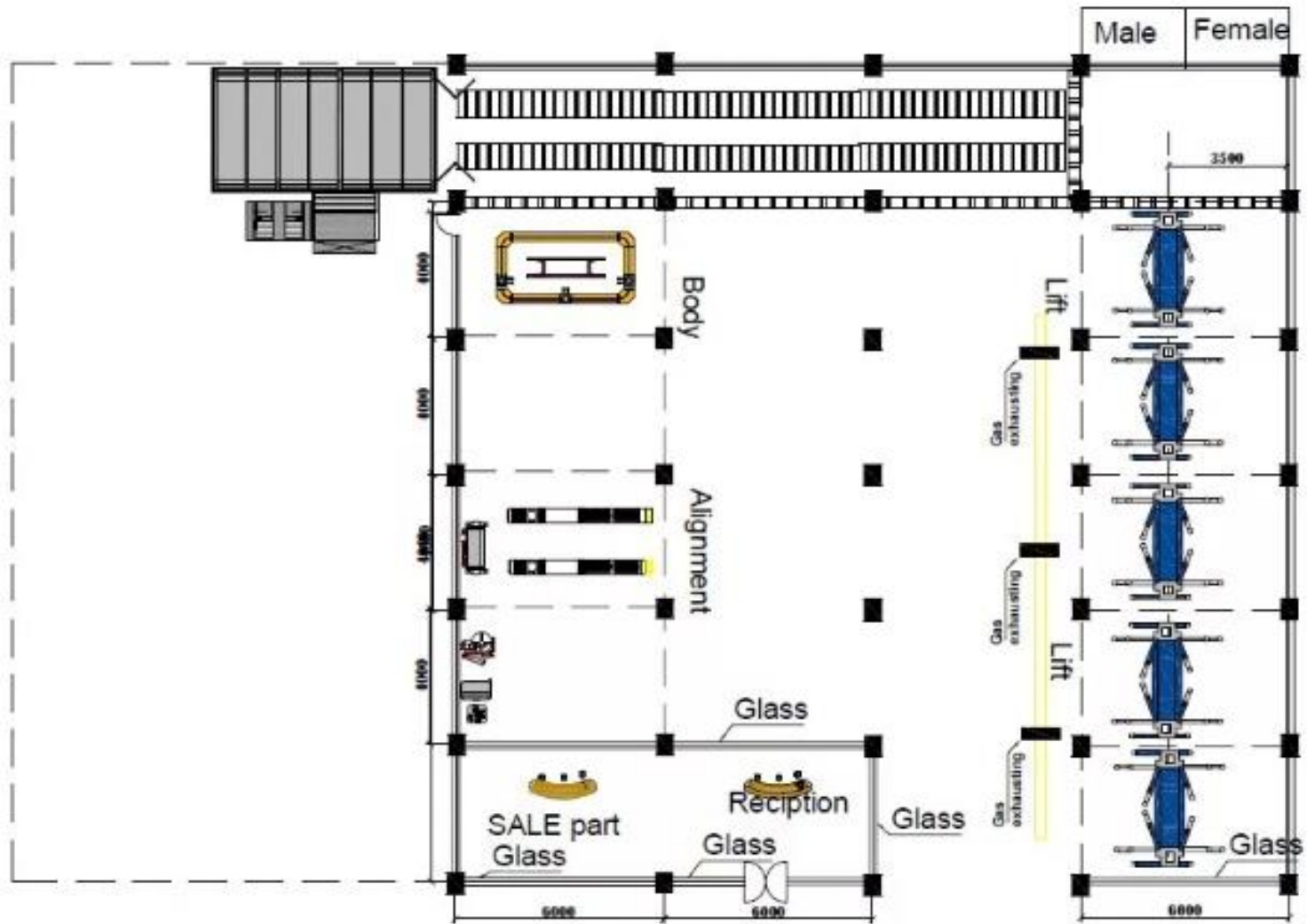
2.4. Infrastructures

The workshop must be of concrete structure. Based on class size of 25 learners, the space requirements for the Training/learning and circulation areas are as follows:

	1.7. Infrastructures	Specification (Length x Width in meters)	Number				
1	Classroom	7x6	1				
2	Workshop/garage	40x30	1				
3	ICT lab	10x8	1				
4	Storage area for consumables	4x6	1				

5	Storage area for tools	9x7	1				
6	Trainers' room	7x6	1				
7	Washroom/changing room	2x4	1				
8	Girl's room	2x4	1				
9	Chemistry Laboratory	10x8	1				
10	Physics Laboratory	10x8	1				
11	EV Battery charging room	3x4	1				
12	Parking	7x10	1				
13	Hazardous materials storage room	2x3	1				
14	Hybrid isolation room	6x10	1				
15	Diagnostic area	7x6	1				

WORKSHOP LAYOUT



N.B:

The occupation of a training classroom is in function of a group of 25 learners for the length of the training package not considering the hours of the internship. The group of trainees is divided into 5 (25 learners per workshop).

The plans of the workshops and classrooms are annexed to the training organization guide.

Total hours of occupation must be equal to the total duration of the training (excluding the duration of the internship).

The percentage of the occupation must equal 100%.

The premises that are not used in learning activities can have hours / occupation. So, we will write 0 for these types of premises.

Reference books:

1. Automotive technology volume principles diagnosis and services fourth edition James D. Haldman, 2012
2. Automotive technology volume a systems approach 5th edition Jack Erjavec, 2009
3. Technology of automotive trade volume 2, H. Girschler, 2011
4. Modern automotive technology Europa Lehmetel 2nd edition, 2011
5. <https://www.aliexpress.com/w/wholesale-AUTOMOBILE.html?spm=a2g0o.home.search.0>
6. Advanced Engine Technology [2 ed.] by HEINZ HEISLER, 2002
7. The Repair of Vehicle Bodies 7th Edition, Andrew Livesey 2019
8. Modern Automotive Technology 7th Edition, James Duffy, The Good heart-Wilcox Company, Inc., 2009. (ISBN: 978-1-59070-956 -6)
9. Auto Body Repair Technology, James E. Duffy (2008, Delmar Cengage Learning) - libgen.lc
10. Automotive body Work and Rust Repair – Matt Joseph 2009
11. The complete guide to Auto body Repair 2nd Edition – Dennis Parks 2015
12. Engineering drawing second edition K. Venkata Reddy, 2008
13. Machine drawing third edition D.H.L. Narayana, Dr. P. Kannaiah, K. Venkata, 1994, 1999, 2006
14. Wilson, R 1990, Engineering Drawing and Design, Edward Arnold, New York.
15. Gupta, R.G. & Prakashan, Satya 1995, A textbook of Engineering Drawing, Khanna Publisher, Delhi.
16. Gopalakrishna, K.R 1998, Engineering Drawing, Subash Publication, Delhi.
17. Morling, K. (2010). Geometric and Engineering Drawing. 3rd ed. Elsevier
18. Shah, M.B. and Rana, C.B. (2009). Engineering Drawing. Darling Kindersley (India) Pvt.Ltd
19. Venkata. K. Reddy. (2008). Textbook of Engineering Drawing. Second Edition

20. Workshop processes practices and materials (by Bruce. Black) 2004, Third Edition
21. Introduction to basic manufacturing process and workshop technology (by Rajadar Singh) 2006
22. https://www.alibaba.com/product-detail/DIXIANG-Wholesale-cheapest-OEM-ODM-14_1601050244212.html?spm=a2700.details.you_may_like.3.23a17476tpspgx
23. https://www.amazon.com/s?k=AUTOMOBILE&i=beauty-intl-ship&crd=3KUMLLLD7EV89&sprefix=automobile%2Cbeauty-intl-ship%2C367&ref=nb_sb_noss_2
24. Introduction to basic manufacturing process and workshop technology (by Rajender singh) 2006
25. Production technology (manufacturing processes, Technology and Automation) by R.K. Jain 2012
26. https://www.beta-tools.co.uk/wp-content/uploads/2022/01/GP_ENG_2022.pdf